

PPA 2

Title

Persistence-Based Evaluation in Non-Objective Computational Systems

Field of the Invention

The invention relates to artificial intelligence, machine learning, optimization systems, and evolutionary computation.

Background

All known learning systems rely on explicit evaluation criteria such as loss functions, fitness functions, reward signals, or human-in-the-loop selection. These impose external goals and distort exploratory behavior.

Biological evolution does not evaluate outcomes against goals. Structures persist or disappear based on compatibility with constraint.

Summary of the Invention

The invention describes computational systems in which:

- No objective function exists
- No scoring, reward, or fitness evaluation is performed
- Selection occurs solely through persistence under transformation

In such systems, continued existence is the sole indicator of viability.

Detailed Description

Patterns are generated intrinsically by interactions within a substrate. Transformations are applied either continuously or discretely. Patterns that destabilize vanish; patterns that remain coherent persist.

Persistence reshapes the future transformation space without explicit memory, scoring, or optimization.

The system does not “select” patterns. It simply permits or forbids continued existence.

Key Novel Claims (Provisional)

- Evaluation via ontological persistence rather than numerical metrics
- Learning without reward, loss, or optimization

- Elimination of external evaluators or selectors
- Replacement of fitness with structural survivability